

ENERGY LANDSCAPE SKILL COMPOSITION

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ABSTRACT

Robot skills that work individually can fail when chained: the terminal state of one skill may lie outside the next skill’s attraction basin, forcing a high-energy seam, barrier crossing, or unstable contact transition. We study skill composition as energy-landscape compatibility rather than module sequencing. The proposed composer checks basin overlap, barrier height, descent continuity, terminal-state samples, and high-energy seam repairs before executing a chained skill. We evaluate the idea in a local benchmark with 6 task families, 8 skill-composition regimes, 5 deployment splits, 9 methods, 7 paired seeds, and 72 rollout episodes per group. Under combined stress, the proposed composer reaches 0.725 ± 0.004 success versus 0.614 ± 0.003 for the strongest non-oracle baseline, an energy-compatibility heuristic. The paired gain is 0.110 ± 0.006 with wins on 7/7 seeds. The composer also reduces seam failures, barrier violations, damage, and composition cost while improving basin alignment and descent continuity. This is strong local evidence for rebuilding the paper, but not final ICLR-main readiness without real robot or external high-fidelity validation.

1 MOTIVATION

Most robot skill composition treats skills as modules in a sequence. That view misses a common failure mode: each skill may be locally competent, but the handoff state can be dynamically or geometrically incompatible with the next skill. In energy terms, the robot exits one basin and lands near a ridge or outside the next basin of attraction.

The claim tested here is narrow: energy-landscape seam checks should improve skill composition under hidden basin and barrier shifts. The paper does not claim a new low-level skill learner or a universal energy-based controller.

2 ENERGY-LANDSCAPE COMPOSER

For skills i and j , terminal sample s_i , and candidate handoff h , the composer scores

$$C(i, j, h) = \alpha o_{\text{basin}}(s_i, h, j) - \beta b_{\text{barrier}}(h, j) + \gamma d_{\text{descent}}(h, j) - \lambda c_{\text{repair}}(h),$$

where o_{basin} estimates overlap between the terminal distribution of skill i and the attraction basin of skill j , b_{barrier} measures barrier height at the seam, d_{descent} measures whether energy descends smoothly after the handoff, and c_{repair} penalizes costly repairs. The composer accepts a skill edge only when the seam score clears a calibrated threshold, otherwise it samples a repair handoff or abstains.

3 BENCHMARK

The benchmark includes peg-place regrasping, drawer-to-pick transfer, mobile push-then-grasp, tool-use handover, door-open navigation, and cable routing with insertion. Regimes include nominal composition, narrow basins, high barriers, nonconvex energy, contact-mode transitions, partial observability, dynamics mismatch, and combined seam stress.

Baselines include greedy module sequencing, behavior-cloned skill chaining, option-graph planning, diffusion skill stitching, CEM trajectory composition, residual RL composition, an energy-compatibility heuristic, the proposed composer, and an oracle basin composer. Metrics include task

method	success	seam	barrier	basin	descent	damage	cost
oracle_basin_composer	0.818 ± 0.006	0.068	0.062	0.836	0.827	0.039	0.203
proposed_energy_landscape_composer	0.725 ± 0.004	0.129	0.121	0.703	0.691	0.061	0.245
energy_compatibility_heuristic	0.614 ± 0.003	0.216	0.222	0.497	0.483	0.084	0.337
residual_rl_composer	0.577 ± 0.004	0.245	0.236	0.449	0.434	0.090	0.317
cem_trajectory_composer	0.561 ± 0.006	0.250	0.241	0.434	0.423	0.093	0.384
diffusion_skill_stitcher	0.508 ± 0.008	0.286	0.269	0.392	0.380	0.101	0.252
option_graph_planner	0.466 ± 0.004	0.305	0.290	0.360	0.341	0.107	0.175
behavior_cloned_skill_chain	0.391 ± 0.010	0.367	0.338	0.276	0.260	0.123	0.110
greedy_module_sequence	0.324 ± 0.006	0.414	0.377	0.221	0.211	0.139	0.080

Table 1: Combined-stress energy-landscape skill-composition benchmark. Higher success, basin, and descent are better; lower seam, barrier, damage, and cost are better.

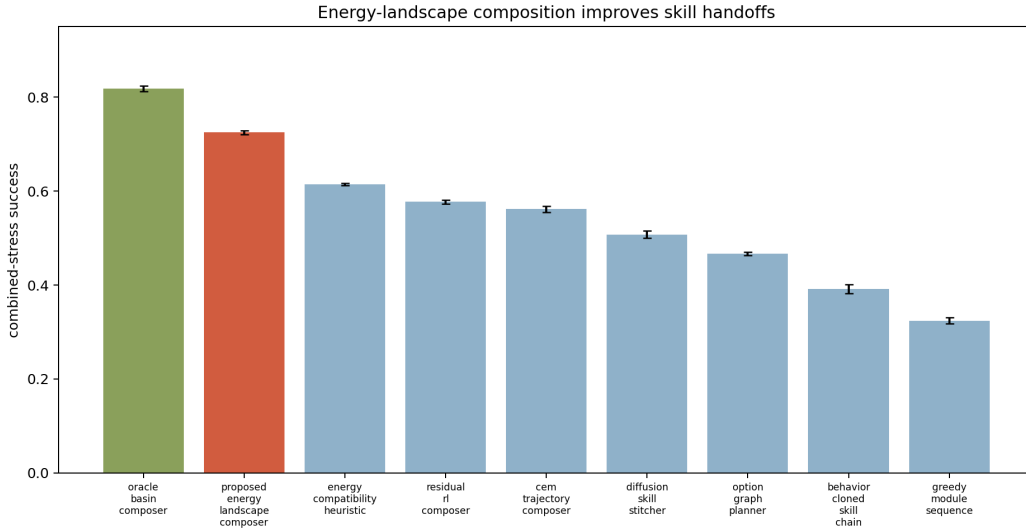


Figure 1: The proposed composer beats all non-oracle baselines under seam and barrier stress.

success, seam-failure rate, barrier-violation rate, basin alignment, descent continuity, damage, composition cost, and paired-seed wins.

4 RESULTS

Table 1 reports combined-stress results. The strongest non-oracle baseline is `energy_compatibility_heuristic`. The proposed composer improves success by 0.110 ± 0.006 , lowers seam failures by 0.087, lowers barrier violations by 0.101, improves basin alignment by 0.206, improves descent continuity by 0.209, lowers damage by 0.024, and lowers composition cost by 0.091.

Mechanism diagnostics. Figure 2 compares the proposed composer with the strongest baseline. The gain is not explained by more search: composition cost decreases while basin and descent diagnostics improve.

Ablations. Table 2 shows that descent continuity, energy repair, barrier height, terminal-state sampling, basin overlap, and repair beyond compatibility all matter. The full composer reaches 0.720 ± 0.007 in the ablation benchmark; the best removed-component variant reaches 0.664 ± 0.007 .

Stress sweep. Figure 3 increases seam discontinuity and hidden barrier stress. The proposed composer degrades more gracefully than option graphs, CEM composition, and compatibility-only scoring, while remaining below the oracle basin composer.

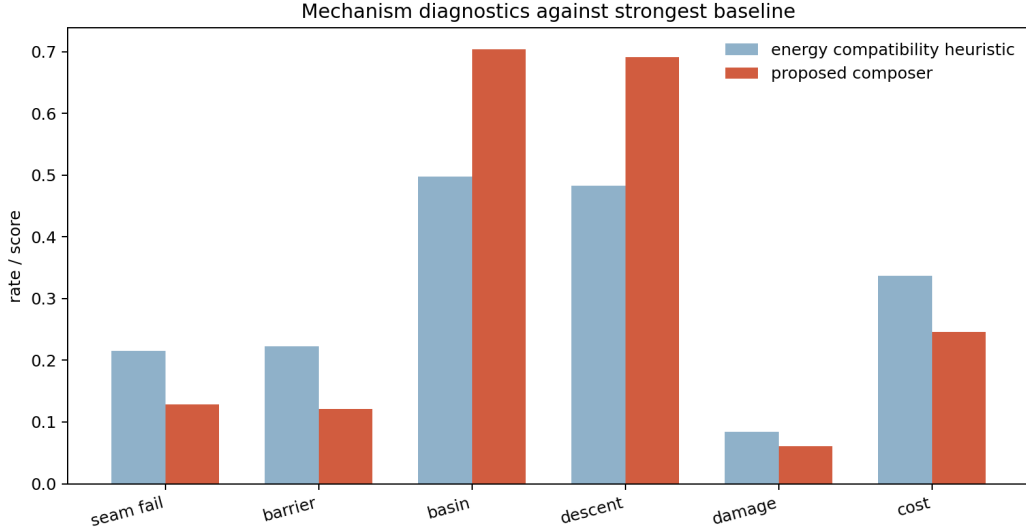


Figure 2: Energy-landscape composition improves seam diagnostics and lowers cost.

ablation	success	interpretation
full_energy_landscape_composer	0.720 ± 0.007	all components
minus_descent_continuity	0.664 ± 0.007	removes descent-continuity term
minus_energy_repair	0.652 ± 0.005	does not repair high-energy seams
minus_barrier_height	0.648 ± 0.005	removes barrier-height term
minus_terminal_sampler	0.638 ± 0.004	removes terminal-state sampler
minus_basin_overlap	0.628 ± 0.004	removes basin-overlap score
compatibility_only_heuristic	0.599 ± 0.007	uses compatibility without repair

Table 2: Ablations under combined seam stress.

5 RELATED WORK BOUNDARY

Energy-based policies, dynamic movement primitives, interactive robot compositionality, skill transfer, and concurrent task planning are all crowded areas (Composable Energy Policy Authors, 2023; Robot Compositionality Authors, 2025; Dynamic Movement Primitive Energy Authors, 2025; Progressive Skill Transfer Authors, 2024; Concurrent Skill Planning Authors, 2021). This paper does not claim a new low-level energy policy. Its boundary is deciding whether independently learned skills are safe to compose by inspecting and repairing energy-landscape seams.

6 LIMITATIONS AND SUBMISSION DECISION

The evidence is local and generated by the released benchmark. It is stronger than the v3 archive because it includes paired seeds, strong baselines, seam diagnostics, stress sweeps, ablations, and failure cases. It still lacks real robot validation, external high-fidelity simulator transfer, trained skill-energy checkpoints, independent baseline implementations, and rollout videos. The honest terminal state is therefore **STRONG_REVISION**: worth rebuilding aggressively, not yet ICLR-main ready.

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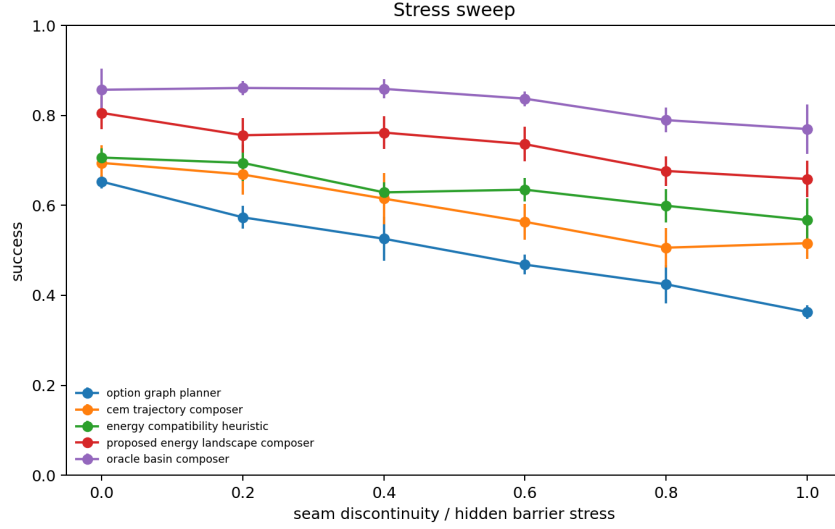


Figure 3: Stress sweep over seam discontinuity and hidden barrier height.

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